SENA: Search Engine with Neural Attention for Long COVID Retrieval

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***Abstract*—Long COVID is a condition associated with persistence or emergence of symptoms weeks or months after acute SARS-CoV-2 infection and has become an important public health concern. Although a rich volume of COVID-19 publications is available, only a small portion of the articles specifically address Long COVID, making efficient retrieval of relevant evidence challenging. Traditional search tools based on keyword matching may fail to capture the evolving terminology and conceptual variations used to describe Long COVID. Consequently, a search engine tool that learns the semantic concept of Long COVID rather than relying solely on keyword matches is essential. In this paper, in a supervised learning classification task, we use titles and abstracts of publications to identify Long COVID literature. We used a publicly available dataset of 552 labeled COVID-related articles to train and evaluate three deep learning approaches, including a Multi-Layer Perceptron (MLP), a Bidirectional Long Short-Term Memory network (BiLSTM), and a transformer classifier based on neural attention. Experimental results indicate that the Search Engine Neural Attention (SENA) model achieved the best overall performance on the hold-out test set, with 0.9 accuracy, 0.9 F1-score, 0.86 precision, and 0.96 recall. In comparison, the MLP and BiLSTM models achieved relatively lower performance in terms of accuracy and F1-score. The high recall value in SENA is particularly valuable for literature retrieval tasks, as it minimizes missed Long COVID studies. These findings demonstrate that contextual learning based on self-attention outperforms traditional baseline models and offer a practical search engine tool for Long COVID research.**

***Index Terms*—Long COVID, semantic search, deep learning, transformer models, BERT, attention mechanism, text mining, information retrieval**

I. Introduction

Since the first reported case of Coronavirus Disease 2019 (COVID-19) in late 2019, the pandemic has generated unprecedented global health [1]. Worldwide, hundreds of millions confirmed infections and millions of deaths have been recorded [2]. Large-scale scientific efforts have produced millions of COVID-19 related publications spanning virology, epidemiology, treatment, vaccination, and public health [3,4]. Although this rapid expansion of the literature has accelerated knowledge discovery, it has also created major challenges in efficiently locating specific information within an increasingly large document space.

One of the most significant post-pandemic clinical concerns is Long COVID, a condition describing persistent, recurring, or new emerging symptoms that may continue for weeks or months after acute SARS-CoV-2 infection [5]. Reported symptoms included fatigue, respiratory symptoms, cardiovascular dysfunction, and as a result, reduced quality of life [6]. Despite growing recognition, Long COVID remains difficult to define consistently because symptom duration thresholds vary across studies, terminology is still evolving, and many early infections were undocumented or asymptomatic [5]. The absence of standardized terminology to describe Long COVID, makes it difficult to identify relevant literature using traditional keyword-based search methods. This challenge is amplified by the rapid expansion of COVID-19 research, while only a relatively small fraction of COVID-related articles specifically address Long COVID [7]. Consequently, researchers may face a delay in locating specific information relevant to Long COVID which is needed to better understand the long-term consequences of infection.

Recent advances in artificial intelligence (AI), natural language processing (NLP), and deep learning have significantly improved biomedical information retrieval by enabling semantic understanding beyond exact keyword matching. Transformer-based language models have become state-of-the-art for text classification, and question-answering tasks [8]. Several AI-driven COVID-19 search systems have demonstrated the potential of these methods. Langnickel et al. introduced preVIEW, a semantic search engine for COVID-19 preprints that incorporated a BERT-based classifier to identify Long COVID articles, reporting an F1-score of approximately 0.91, demonstrating high precision and recall [7]. Furthermore, Esteva et al. developed an AI-powered search engine that combines BERT-based retrieval with keyword-based methods to identify relevant COVID-19 research. Their system incorporates a question-answering framework using transformer models and GPT-2, allowing more interactive and user-focused information extraction [9]. Another notable system is the AWS CORD-19 Search platform, which integrates neural search techniques with knowledge graph approaches in tasks including document ranking, passage retrieval, and question-answering over COVID-19 literature [10]. Additionally, the COVID-19 Search Engine (CO-SE) employs a transformer-based architecture to extract precise answers from scientific literature achieving approximately 71% Exact Match and 79% Semantic Answer Similarity, indicating its capability to retrieve relevant information from large datasets [11].

Despite these advances, there remains a need to systemically evaluate modern deep learning architectures for the identification of Long COVID literature using publicly available labeled datasets. In particular, it is valuable to compare lightweight neural baselines with more advanced transformer models to better understand the trade-off between simplicity, computational cost, and predictive performance.

Motivated by the existing challenges in Long COVID literature identification as well as the lack of a pipeline to compare the performance of different algorithms, the goal of this study is to leverage AI in text mining and propose a search engine tool that identifies Long COVID publications from their titles and abstracts. We utilized a labeled dataset available on Hugging Face [12], originally published by Langnickel et al. [7], which contains 552 COVID-19-related articles annotated for long COVID relevance. Furthermore, we compare the performance of three modeling architectures, including an MLP baseline, a BiLSTM, and the proposed search engine with neural attention (SENA) tool for Long COVID document classification. By improving the retrieval of Long COVID publications, this work aims to provide insights of using different deep learning classifiers in an NLP task, as well as improving search accuracy and reducing the time required for researchers to identify relevant studies, ultimately supporting faster scientific discovery and better understanding of Long COVID.

II. Methods

In this project, we will solve a binary classification problem for identifying Long COVID-related publications from all COVID articles. The objective is to learn the mapping function:

$$F(x) \rightarrow y \tag{1}$$

Where:

- x represents the input data with text format including title and abstract of paper.
- $y \in \{0, 1\}$ is the output label, $y = 1$ represents the Long COIVD-related articles, and $y = 0$ represents the non-Long COVID publications.

A. Dataset Description

We utilize a publicly available Hugging Face dataset known as "long-covid-classification-data" [12], which is originally derived from the published literature of Langnickel et al. where they introduced the preVIEW semantic search system [7]. The dataset consisted of 552 articles, split into 414 training, and 138 test sets. For each sample, we have inputs as text, and binary label. As shown in Fig. 1, the dataset is balanced in terms of class distribution, ensuring equal representation of positive and negative examples in both train and test splits. In addition, no missing values were observed.

B. Text Preprocessing and Tokenization

Since the inputs are all text, in order to use them with machine learning classifiers, all needed to be tokenized first. We tokenized all documents using the BERT base uncased tokenizer from the transformers module in python. As a result, each input data was converted into token representations compatible with transformer and neural architectures. The preprocessing pipeline includes the BERT uncased tokenizer for tokenization, sequence truncation at 512 tokens, padding to maximum length, and attention mask generation to distinguish padding from real tokens. Formally, each input document is transformed into input ids and attention mask. Finally, the tokenized dataset is converted into PyTorch tensor format to be compatible with PyTorch and batching with "DataLoader" module.

C. Model Architectures

To evaluate the effectiveness of different neural architectures for biomedical text classification, three neural network models were implemented: a baseline MLP model, an LSTM sequence model, and finally a transformer-based classifier named SENA. We utilized a gird search approach to identify optimal hyper-parameter values including Batch sizes (8, 16, 32), Learning rates (1e-3, 2e-5, 2e-7), Hidden layers (1, 3, 5), number of neurons (64, 128, 256), dropout (0, 0.3, 0.5), optimizers (Adam, AdamW) and number of epochs (1 to 10). Here we provide the model architectures for the models with highest test performance based on accuracy metric:

1) Multi-Layer Perceptron (MLP) Baseline: The MLP model serves as the baseline model. This model consists of:

- Embedding layer which maps token IDs to dense vectors. This layer was initialized with the BERT vocabulary size tokenizer.
- Mean pooling over token embeddings (masked by attention mask).
- Fully connected hidden layer with 64 neurons and ReLU activation function.
- Output layer producing logits for two classes.

Also, the loss function for the MLP model is the Cross Entropy Loss function, and Adam optimizer is used for optimization. The learning rate is set to 1e-3.

2) Bidirectional LSTM (BiLSTM): The LSTM model captures sequential dependencies in text using a recurrent neural network architecture. The model consists of:

- The token embedding layer (similar to the MLP model).
- Bidirectional LSTM encoder.
- Dropout regularization layer with rate = 0.5.
- Fully connected classification head.

Moreover, LSTM hidden layer consists of 128 neurons, and identified that Adam with learning rate of 1e-3 has the best performance. We observed that using higher learning rates will lead to overfitting of the model by increasing the train accuracy without increasing test accuracy. In addition, 9 epochs were identified as optimal. Finally, BiLSTM compared to Unidirectional LSTM, provided lower loss and higher accuracy for each epoch on the test set.

3) Transformer-Based Model (SENA): The proposed model, referred to as SENA, is based on the pretrained BERT (bert-base-uncased) architecture. SENA consists of:

- Pretrained BERT encoder.
- Extraction of the CLS token representation.
- Dropout layer with rate 0.5 for regularization.
- Linear classification head for binary classification.
- AdamW optimizer with learning rate 2e-5 and Cross Entropy Loss.

Also, for faster training, we used the "Data Collator With Padding" from transformers library to dynamically pad per batch instead of pad all sequences to a fixed length, which we used previously in MLP and LSTM models. Unlike the baseline models, SENA leverages the self-attention mechanism, allowing each token to attend to all other tokens in the sequence. This represents stronger learning in a context-based format.

D. Evaluation Metrics

All models were trained on the training set and evaluated on the test set. Model performance evaluated using the following classification metrics:

- Accuracy
- Precision
- Recall (Sensitivity)
- F1-score

Given the biomedical nature of this paper, recall is considered an important parameter, as missing Long COVID-related publications (false negatives), can lead to inadequate performance of the search system. Also, F1-score were used in the pipeline to provide an opportunity for evaluation of models on imbalanced data. It is important to consider the F1-score over accuracy if we face imbalanced datasets.

Furthermore, all experiments were implemented in PyTorch. Dataset loaded using "datasets" from Hugging Face, and batching and padding adopted for each model type. The train and test splits were according to the original Hugging Face configuration. Also, for transformer-based training, computations were accelerated using GPU and CUDA when available.

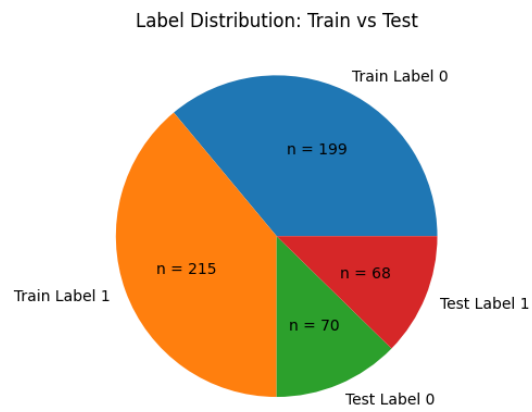

Fig. 1. Distribution of Long COVID and non-long COVID papers across the training and test splits.

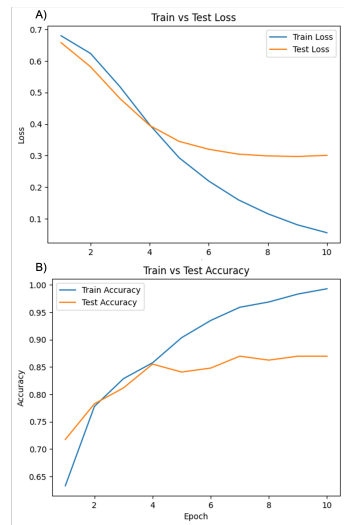

Fig. 2. Performance of MLP model over train and test splits based on A) Loss Function B) Accuracy.

III. Results

In this paper, we evaluated the effectiveness of different neural network architectures for Long COVID document classification. Raw text inputs were first tokenized. The processed dataset was then loaded into PyTorch "Data Loaders" for mini-batch training and evaluation. Three models were compared under the same dataset split to train and test sets: 1) MLP as the baseline neural classifier; 2) BiLSTM as the sequential learning model; and 3) SENA as the transformer-based model. All models were evaluated on the hold-out test set using Accuracy, Precision, Recall, and F1-score.

A. MLP Baseline

The MLP baseline showed steady optimization progress over ten epochs. As shown in Fig. 2, the training loss decreased from 0.68 in the first epoch to 0.06 in the final epoch, and the training accuracy increased consistently from 0.63 to 0.99. In terms of model evaluation, the test loss decreased from 0.66 to 0.30, and the accuracy improved from 0.72 to 0.87.

The decrease in training loss and simultaneous increase in training accuracy indicate that the baseline model had a successful convergence and learning task. On the other hand, the substantial gap between near a perfect training accuracy and lower test accuracy suggests that the model faces overfitting during the last epochs.

Overall, the MLP model demonstrates that even simple neural architectures can learn meaningful patterns from biomedical text when token embeddings are used. However, this model relies solely on pooled embeddings without explicit contextual modeling, and as a result, its ability to generalize remains limited compared with transformer-based methods.

B. Bidirectional LSTM

As shown in Fig. 3, the LSTM model achieved rapid improvements during early training epochs, reaching 0.84 training and 0.82 test accuracies by epoch 3. By increasing the number of epochs, training accuracy continued increasing to 0.99, however, the test accuracy fluctuated between 0.80 and 0.85. This divergence between train and test performance indicates that the LSTM model faces a stronger overfitting problem than the observed MLP model. Although BiLSTM captures sequential word order and local dependencies, it appears less robust on this relatively small dataset. Despite using a dropout rate of 0.5 in the LSTM model and not using dropout in the MLP, the LSTM model likely memorized training patterns with less generalizability to unseen documents. This suggests that more advanced architectures like LSTMs may require larger datasets for training to achieve an optimal performance.

C. SENA (Transformer-Based Neural Architecture)

Unlike two previous models, the advanced search engine tool based on neural attention mechanism converged efficiently within only 4 epochs. As shown in Fig. 4, we observe the lowest difference in loss between train and test splits on epoch 3. The test accuracy reached 0.72 on the first epoch and increased to 0.87 on epoch two. Also training accuracy improved from 0.63 on the first epoch to 0.83 on the second. We stopped at epoch 4 because we observed overfitting (reducing train loss and increasing test loss) after this epoch.

The rapid convergence of SENA with just a few epochs reflects the benefit of transfer learning from pretrained language models. Unlike MLP and LSTMs, SENA based on self-attention mechanism considers the contribution of each word in a context format and captures the contextual relationship between words, providing an opportunity to achieve the target goal of classification with fewer epochs.

As shown in Fig. 5, we compared the performance of final models (i.e., final trained model on the last epoch) using different evaluation metrics. This reveals several key insights for us:

- In terms of document classification, a transformer learning model achieves the best overall performance compared to MLP and LSTMs, obtaining the highest Accuracy and F1-score in SENA.
- SENA produced the highest recall (0.96), indicating superior sensitivity for detecting Long COVID literature. This is very important to us, as in real-world application we face a very imbalanced document sets with few positive cases. Not losing any positive case is more important than a false positive classification. As a result, recall is a particularly important metric.
- MLP performed competitively, outperforming the LSTM despite its simpler architecture.
- LSTM yielded the poorest generalization by obtaining the lowest F1-score, suggesting sensitivity to limited data.
- MLP offers lower computational cost and competitive accuracy with SENA.

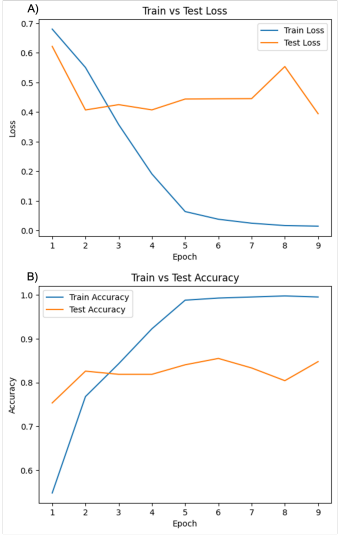

Fig. 3. Performance of BiLSTM model over train and test splits based on A) Loss Function B) Accuracy.

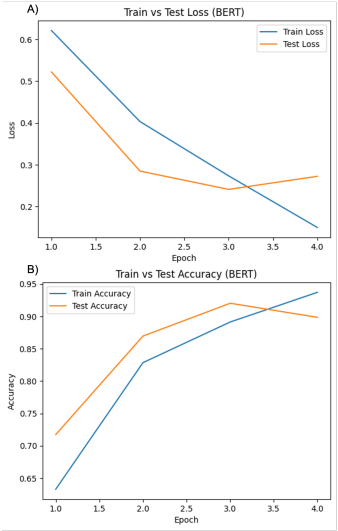

Fig. 4. Performance of SENA model over train and test splits based on A) Loss Function B) Accuracy.

- Moreover, considering all epochs, the average performance of SENA is higher than both MLP and BiLSTM in terms of test accuracy (Fig. 6).

Overall, all these results suggest that the transformer-based model is the most effective approach for Long COVID literature classification according to different evaluation metrics. Although simpler baselines have lower computational costs, the transformer model remains superior in both generalization and sensitivity. Based on F1-score values, by increasing the number of samples or in an imbalanced dataset, we should expect that SENA significantly outperforms MLP and LSTM models. These findings support the use of modern transformer architectures in biomedical semantic search.

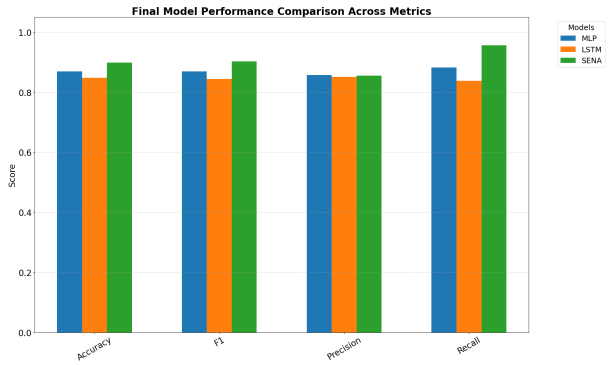

Fig. 5. Evaluation of final models using different metrics.

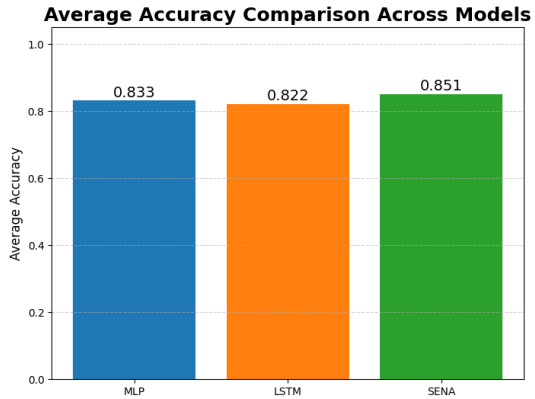

Fig. 6. Average accuracy of each model across the number of epochs.

IV. Conclusion

This study addressed the challenge of identifying Long COVID-related publications within a wide range of COVID-19 literature. Because of the small fraction of COVID-related articles focused on Long COVID, and inconsistent terminology to define Long COVID, traditional keyword search may fail to capture relevant studies. To overcome this limitation, in a supervised binary text classification task using publication titles and abstracts, we designed a search engine tool based on semantic search that learns the concept of Long COVID rather than relying solely on exact keyword matching.

Using a publicly available dataset from Hugging Face, we designed and compared three deep learning architectures: a MLP baseline, a BiLSTM, and the proposed transformer-based SENA model built on pretrained BERT. The results demonstrated that SENA achieved the best overall performance on the independent set, obtaining 0.90 accuracy, 0.90 F1-score, 0.86 precision, and 0.96 recall. In comparison, the MLP achieved 0.87 accuracy and 0.87 F1-score, and the BiLSTM achieved 0.85 accuracy and 0.84 F1-score (Fig. 5).

These findings indicate that learning in a context format through self-attention provides measurable advantages over traditional perceptron and recurrent neural models. In particular, the high recall achieved by SENA is especially important for real-world retrieval systems, where missing relevant publications can delay search and scientific understanding. Although the MLP baseline performed competitively on this dataset, SENA offered the strongest balance between sensitivity, generalization, and overall predictive performance.

An important limitation of this study is the relatively small and balanced dataset size. Although this setting enabled controlled benchmarking, it does not fully reflect real-world literature environments where Long COVID papers are substantially fewer than general COVID publications. Future work should therefore evaluate these architectures on larger, more diverse, and imbalanced datasets. Additional directions include external validation on a separate test set, implement SENA in a multi-class classification task, and rank papers based on relevance score, which all can enhance model robustness and generalizability.

In conclusion, this study successfully met its objective of evaluating AI-based semantic text classification methods for Long COVID literature retrieval. The results support the proposed model SENA, as a practical and effective foundation for biomedical search systems and accelerate understanding of the Long COVID condition.

V. Data and Code Availability

Dataset is publicly available through Hugging Face [12]. Also, all codes are available at GitHub repository: https://github.com/Sina2285/stats507-coursework/tree/main/Final-Project/Final

References

[1] Li, Q. et al. Early Transmission Dynamics in Wuhan, China, of Novel Coronavirus–Infected Pneumonia. N. Engl. J. Med. 382, 1199–1207 (2020).

[2] Barajas, A., Riquelme-Alacid, G., Vera-Montecinos, A. & Ramos, B. Cognition, Cytokines, Blood–Brain Barrier, and Beyond in COVID-19: A Narrative Review. Int. J. Mol. Sci. 27, (2026).

[3] http://datastudio.google.com/reporting/70fdce85-1b53-4605-8698-839b66c8010b/page/JwhJB?feature=opengraph.

[4] Else, H. How a torrent of COVID science changed research publishing — in seven charts. Nat. Commun. 588, 553–553.

[5] Raveendran, A. V., Jayadevan, R. & Sashidharan, S. Long COVID: An overview. Diabetes Metab. Syndr. Clin. Res. Rev. 15, 869–875 (2021).

[6] Merhavy, Z. I. et al. Long COVID: A Comprehensive Overview of the Signs and Symptoms across Multiple Organ Systems. Korean J. Fam. Med. 45, 305–316 (2024).

[7] Langnickel, L., Darms, J., Heldt, K., Ducks, D. & Fluck, J. Continuous development of the semantic search engine preVIEW: from COVID-19 to long COVID. Database 2022, baac048 (2022).

[8] Terzic, I., Vrljic, B., Mlinaric, D., Mestrovic, A. & Boticki, I. AI-Driven Systematic Reviews: Transformer-Based Approaches to Text Classification. in Proceedings of the 2025 2nd International Conference on Modeling, Natural Language Processing and Machine Learning 615–622 (Association for Computing Machinery, New York, NY, USA, 2025). doi:10.1145/3757110.3757213.

[9] Esteva, A. et al. CO-Search: COVID-19 Information Retrieval with Semantic Search, Question Answering, and Abstractive Summarization. arXiv.org https://arxiv.org/abs/2006.09595v1 (2020).

[10] Bhatia, P. et al. AWS CORD-19 Search: A Neural Search Engine for COVID-19 Literature. in AI for Disease Surveillance and Pandemic Intelligence: Intelligent Disease Detection in Action. 131–145 (Springer International Publishing, Cham, 2022).

[11] Raza, S. A COVID-19 Search Engine (CO-SE) with Transformer-based architecture. Healthc. Anal. N. Y. N 2, 100068 (2022).

[12] llangnickel/long-covid-classification-data · Datasets at Hugging Face. https://huggingface.co/datasets/llangnickel/long-covid-classification-data